Day 03

Regression with `scikit-learn`

MSU REU Machine Learning Short Course

Classification vs Regression

Same pipeline. Different output.

	Classification	Regression
Output	A category (STAR / QSO)	A number (z-band magnitude)
Loss	Misclassification rate	Mean Squared Error
Evaluation	Confusion matrix, F1	MSE, R^2, residuals
Example	"Is this a star?"	"How bright is this star?"

Linear Regression

The model: $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$

The coefficients β are learned from training data by minimizing MSE.

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

Same API as KNN. That's the point of scikit-learn.

How good is your fit? MSE and R^2

MSE — average squared distance between predicted and actual values. Lower is better.

R^2 — fraction of variance explained by the model. 1.0 is perfect; 0.0 is no better than predicting the mean.

```
from sklearn.metrics import mean_squared_error, r2_score


mse = mean_squared_error(y_test, y_pred)
r2  = r2_score(y_test, y_pred)
```

$R^2 = 0.95$ sounds great. But always look at the plots.

Always visualize — Anscombe's Quartet

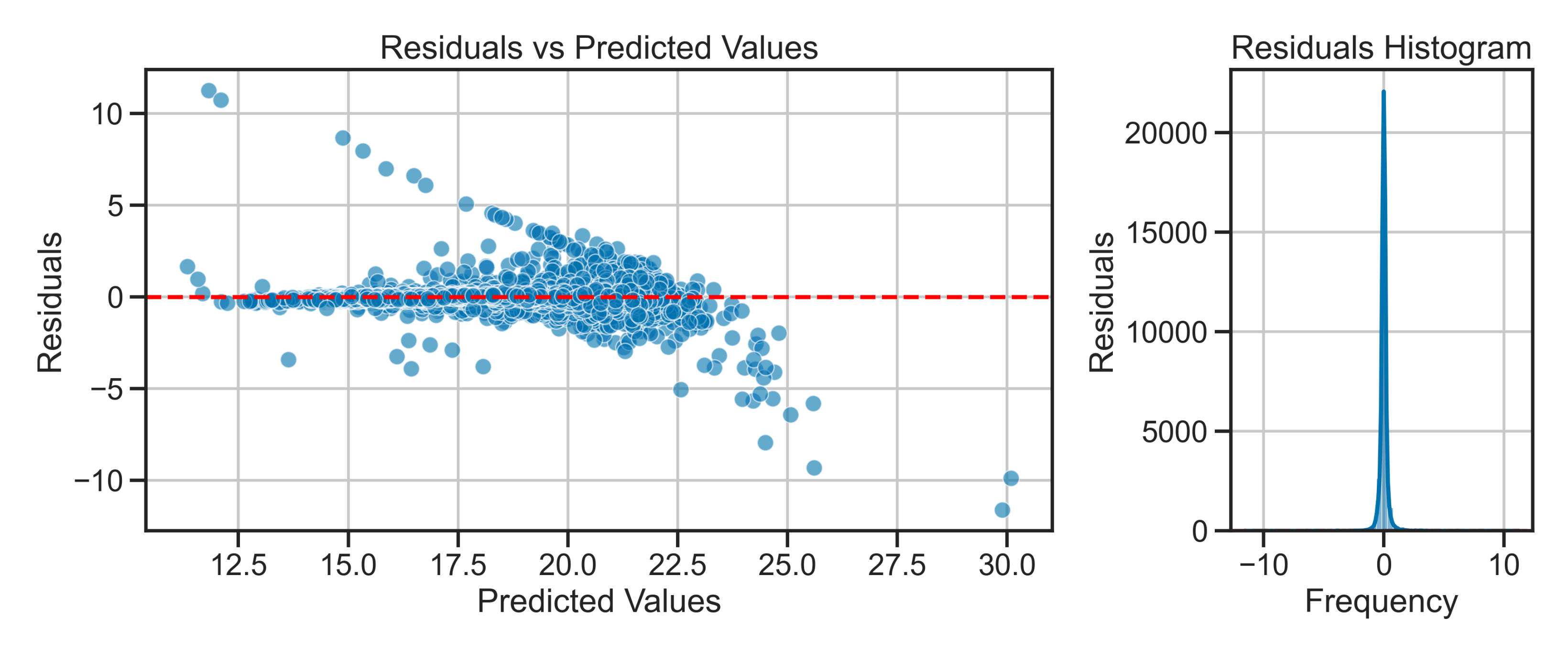
Four datasets. Identical means, variances, correlations, and R^2 values. Completely different patterns.

> Identical statistics **do not** mean identical data.
>
> See: [The Importance of Visualization]

The lesson: before trusting any metric, make the plots.

Three diagnostic plots

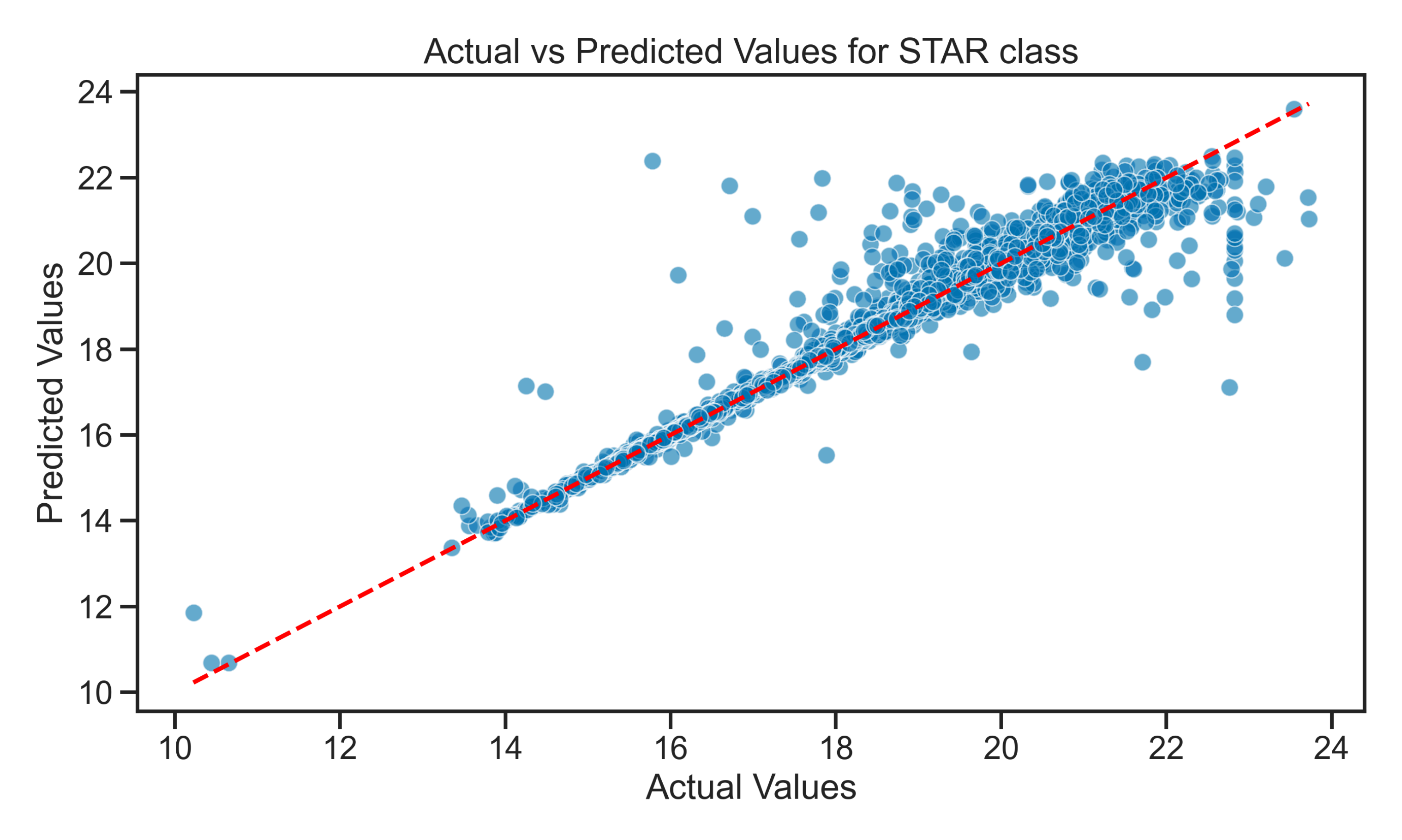
Always make these: residuals, predicted vs actual, and the fit itself.



Residuals should be centered on zero with no structure. If they fan out or curve — your model is missing something.

When linear isn't enough — Random Forest

Same data, different model. The sklearn API doesn't change.




```
from sklearn.ensemble import RandomForestRegressor
```

Today's Activity

Work through **Activity 03: Regression with scikit-learn**

1. Predict z-band magnitude from other photometric bands
2. Evaluate with MSE, R^2, and diagnostic plots
3. Add redshift — does it help regression the way it helped classification?
4. Try per-class models (STAR, GALAXY, QSO separately)
5. Compare Linear Regression vs Random Forest

> **Note:** [The Importance of Visualization]